Assignment 6

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("insurance.csv")
df.head()
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1.0	Male	NaN	Software Engineer	6.1	6.0	42.0	6.0	Overweight	126/83	77.0	4200.0	NaN
1	2.0	Male	28.0	Doctor	6.2	6.0	60.0	8.0	Normal	125/80	75.0	10000.0	NaN
2	3.0	Male	28.0	Doctor	6.2	6.0	60.0	8.0	Normal	125/80	75.0	10000.0	NaN
3	4.0	Male	28.0	Sales Representative	5.9	4.0	30.0	8.0	Obese	140/90	85.0	3000.0	Sleep Apnea
4	5.0	Male	28.0	Sales Representative	5.9	4.0	30.0	8.0	Obese	140/90	85.0	NaN	Sleep Apnea

Next steps: Generate code with df | New interactive sheet

Understand the Data

```
df.shape
df.columns
df.info()
```

Show hidden output

```
df.describe()
```

Show hidden output

```
df.isnull().sum()
```

	0
Person ID	3
Gender	2
Age	2
Occupation	2
Sleep Duration	2
Quality of Sleep	1
Physical Activity Level	3
Stress Level	3
BMI Category	2
Blood Pressure	5
Heart Rate	2
Daily Steps	2
Sleep Disorder	69

dtype: int64

```
df.drop_duplicates(inplace=True)
```

```
# Remove duplicates
df.drop_duplicates(inplace=True)

# Fill missing values (if any)
df.fillna(df.mean(numeric_only=True), inplace=True)
```

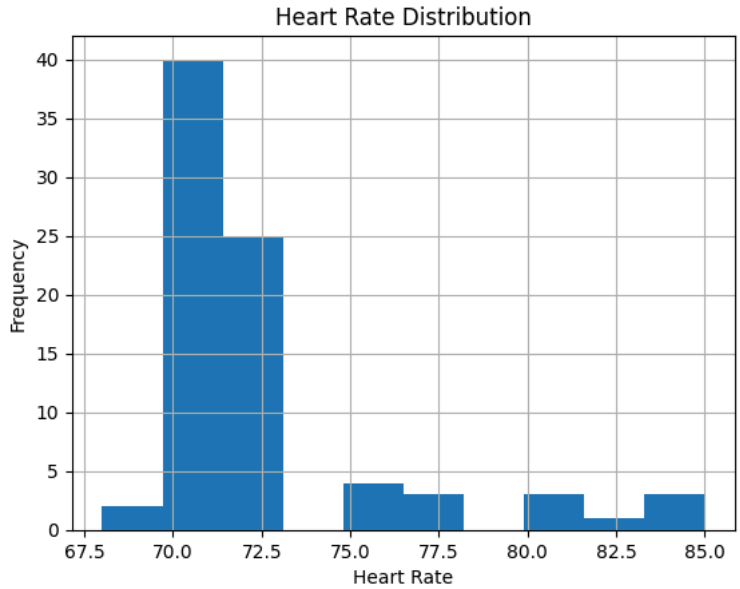
```
# The 'rating' column is categorical, so .mean() cannot be applied.
# Also, after df.dropna(), there are no missing values in this column, so fillna is not needed.
```

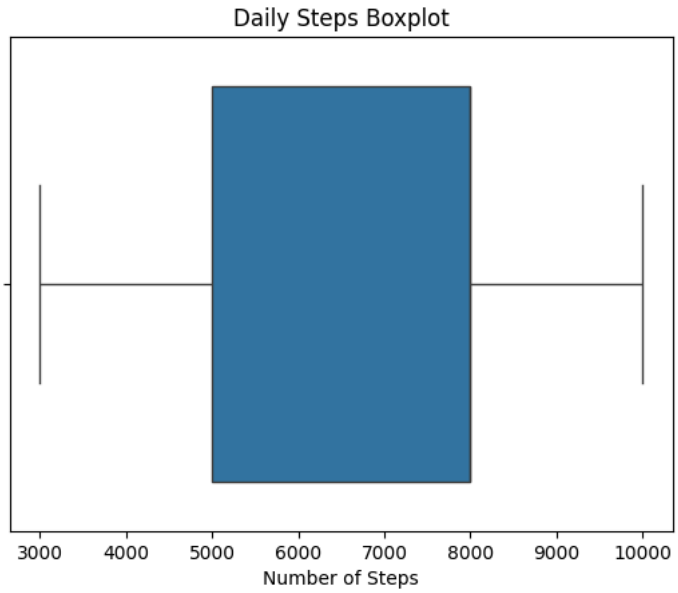
for visulization

```
import matplotlib.pyplot as plt
import seaborn as sns

# Histogram for Age
df['Heart Rate'].hist()
plt.title("Heart Rate Distribution")
plt.xlabel("Heart Rate")
plt.ylabel("Frequency")
plt.show()

# Boxplot for Daily Steps
sns.boxplot(x=df['Daily Steps'])
plt.title("Daily Steps Boxplot")
plt.xlabel("Number of Steps")
plt.show()
```

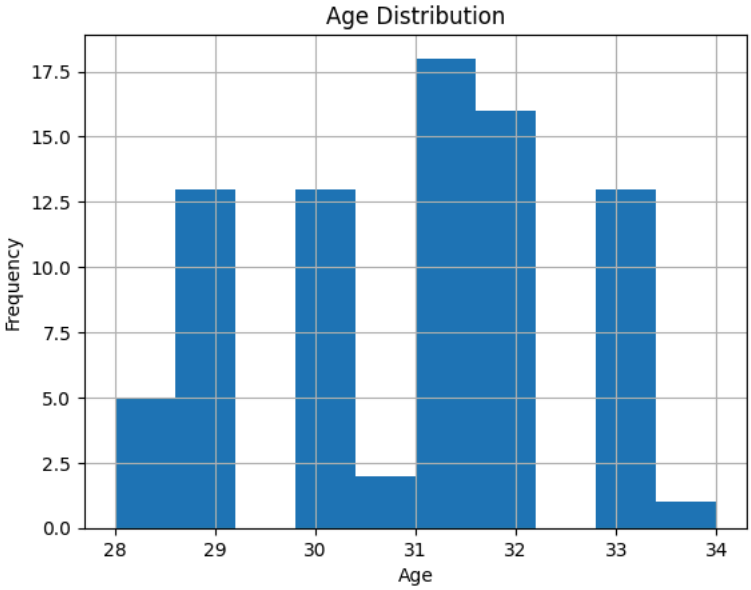


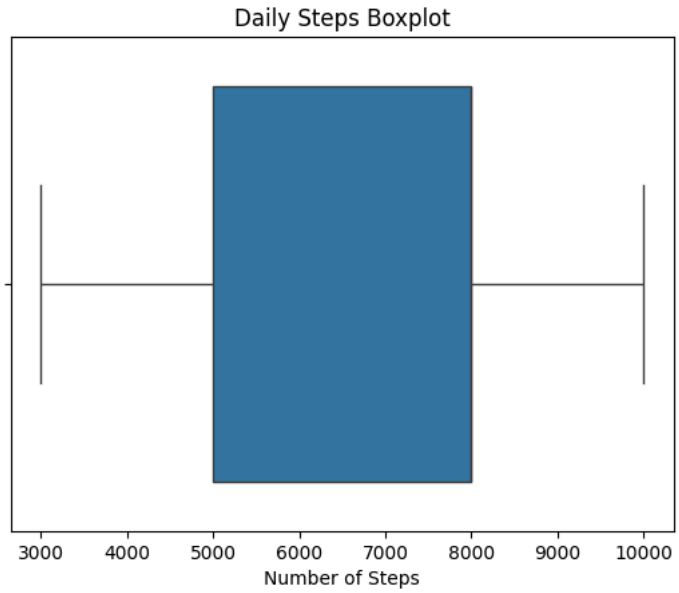



```
# Histogram for Age (drop missing values)
df['Age'].dropna().hist()
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()

# Boxplot for Daily Steps (drop missing values)
sns.boxplot(x=df['Daily Steps'].dropna())
```

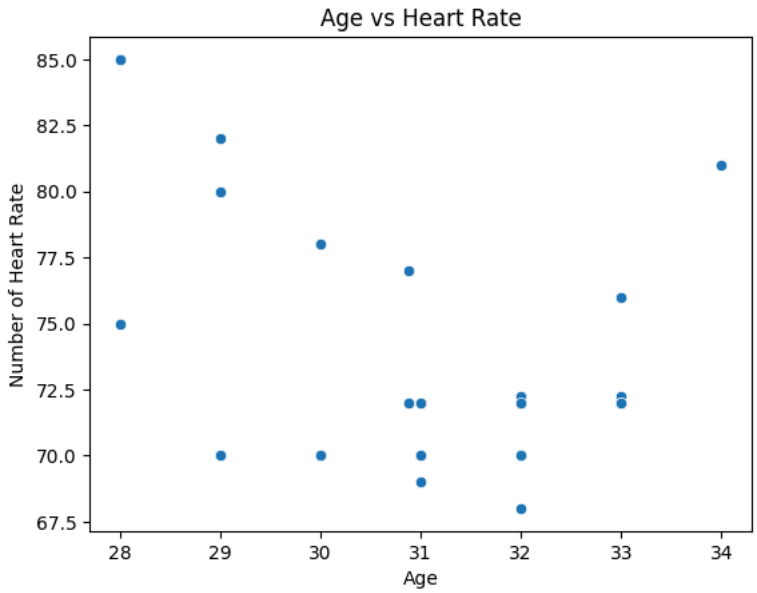
```
plt.title("Daily Steps Boxplot")
plt.xlabel("Number of Steps")
plt.show()
```

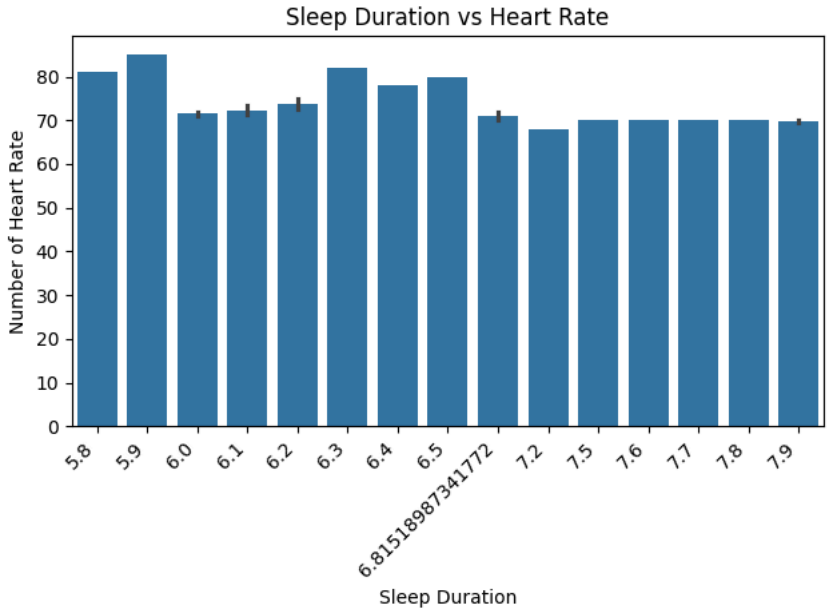





```
sns.scatterplot(x='Age', y='Heart Rate', data=df)
plt.title("Age vs Heart Rate")
plt.xlabel("Age")
plt.ylabel("Number of Heart Rate")
plt.show()


sns.barplot(x='Sleep Duration', y='Heart Rate', data=df)
plt.title("Sleep Duration vs Heart Rate")
plt.xlabel("Sleep Duration")
plt.ylabel("Number of Heart Rate")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()#bivariate
```

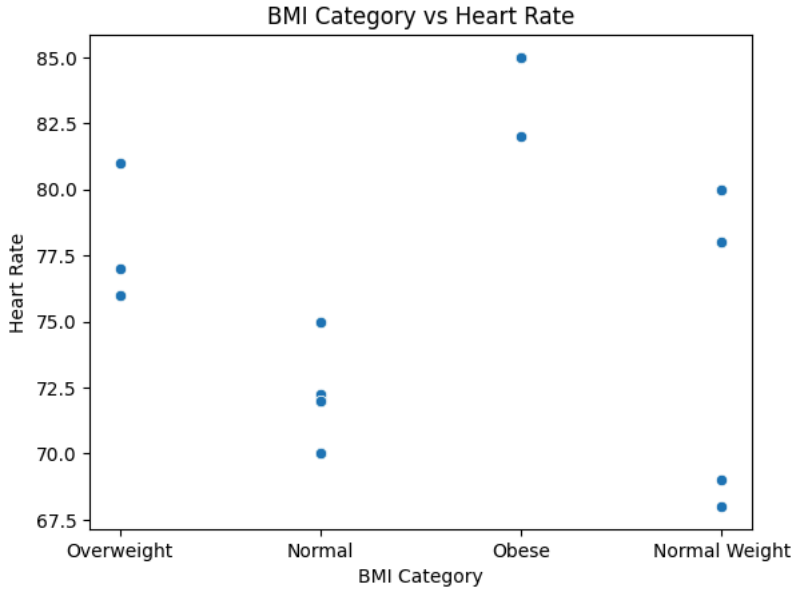


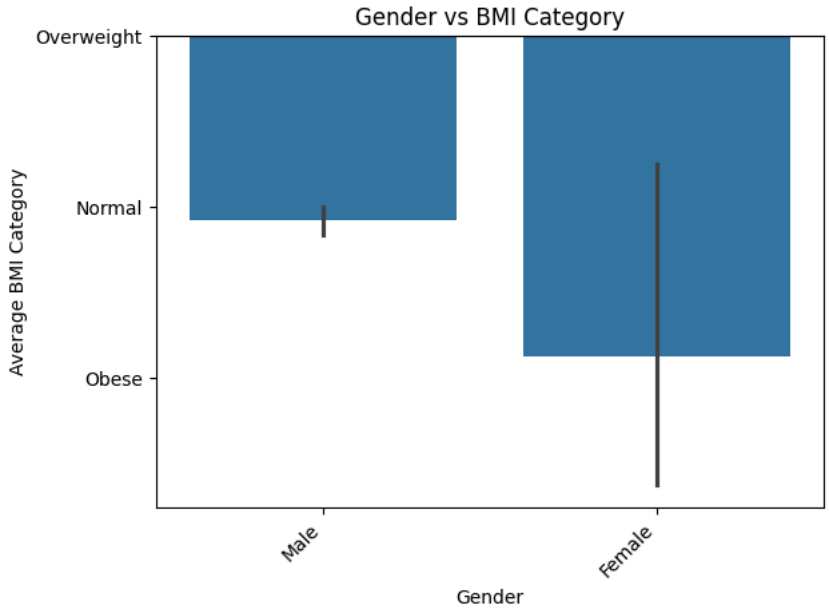



```
import seaborn as sns
import matplotlib.pyplot as plt

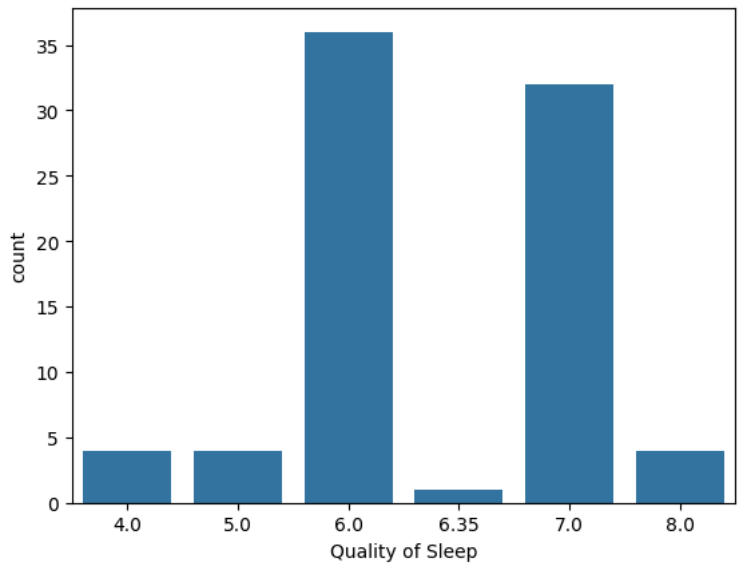
# Example: BMI vs Heart Rate
sns.scatterplot(x='BMI Category', y='Heart Rate', data=df)
plt.title("BMI Category vs Heart Rate")
plt.xlabel("BMI Category")
plt.ylabel("Heart Rate")
plt.show()

# Example: Gender vs BMI
sns.barplot(x='Gender', y='BMI Category', data=df)
plt.title("Gender vs BMI Category")
plt.xlabel("Gender")
plt.ylabel("Average BMI Category")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```






```
sns.countplot(x='Quality of Sleep', data=df)
plt.show()#categories
```




```
plt.figure(figsize=(12, 7))
sns.stripplot(
    x='Gender',
    y='Daily Steps',
    data=df
```

```
    data=df,
    jitter=True,
    hue='Gender',
    palette='viridis',
    legend=False
)

plt.title('Daily Steps Distribution by Gender')
plt.tight_layout()
plt.show()
```

